VAMSI KRISHNA SAMAVEDAM

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Summary

Data Engineer with 2+ years of experience as a Salesforce Developer and MS Computer Science candidate. Skilled in Python and SQL for building and optimizing ETL/ELT pipelines using Azure Data Factory, ADLS Gen2, Databricks, and Synapse Analytics. Strong background in integrating and migrating data across diverse sources while enforcing data quality and compliance. Proven track record streamlining case management and automating workflows to boost data reliability.

Technical Skills

- **Languages:** Python, SQL, C++, Java
- **Data Engineering:** ETL/ELT, Data Pipelines, Data Warehousing, Data Modeling, Dimensional Modeling, star schema, Incremental Loads, Data Quality, Data Validation, Data Migration, Azure Pipelines
- **Cloud & Big Data:** Azure Data Factory, ADLS Gen2, Azure Synapse, Azure SQL Managed Instance, Databricks, Spark
- **Databases & BI:** PostgreSQL, SQL Server, Power BI
- **Salesforce:** Salesforce CRM, Apex, Triggers, SOQL, SOSL, Flows, Validation Rules, Approval Processes, Lightning Components, Data Loader, Profiles, Permission Sets, Role Hierarchy, Sharing Rules
- **Machine Learning:** Anomaly Detection, Autoencoders, Neural Networks, Scikit-learn, Statistical Modeling
- **Tools:** Git, Agile, Matplotlib, Power BI, Seaborn, Excel, Linux

Professional Experience

Accenture | ***Advanced Associate Software Engineer - Salesforce Developer*** **Jan 2023 - Sep 2024**

- Developed and maintained Salesforce solutions with Apex, triggers, SOQL/SOSL, Flows, validation rules, and Lightning components, enabling enterprise CRM workflows, reducing manual data-entry errors, and integrating into broader ELT processes
- Built backend business logic and automation workflows in Apex and Flow for case lifecycle management, routing, and approvals, which streamlined case processing and cut handling time
- Wrote and optimized SOQL queries in the Developer Console to retrieve, validate, and troubleshoot Salesforce data, improving reporting speed and simplifying debugging for business processes
- Performed data migration and validation using Salesforce Data Loader and custom Python scripts on Linux environments, ensuring clean and consistent CRM data across Salesforce objects
- Implemented data quality controls with validation rules, required-field checks, duplicate-prevention logic, and approval workflows within ELT frameworks, reducing duplicate records and enhancing data reliability
- Configured secure data access by setting up Profiles, Permission Sets, Role Hierarchy, Sharing Rules, and Organization-Wide Defaults, which ensured that only authorized users could view sensitive records and improved compliance with data-governance policies
- Participated in client meetings to gather requirements, clarify issues, and provide status updates, then recorded the outcomes in Jira and created detailed technical specifications, enabling the development team to deliver solutions that met business needs on schedule
- Debugged production issues on Linux servers using logs, test cases, and root-cause analysis, and coordinated with QA and business teams to validate fixes before deployment, resulting in faster incident resolution and preventing further service disruptions

Technical Projects

Kitsune Network Intrusion Detection System | Python, Autoencoders, Anomaly Detection **Jan 2026 - Apr 2026**

Kitsune: Network Intrusion Detection System

- Built a real-time network intrusion detection system using ensemble autoencoders (Kitsune), analyzing 700K+ packets and improving anomaly detection accuracy while reducing false positives through adaptive thresholding and drift handling.
- Designed an adaptive thresholding mechanism (mean + z·std) with drift handling to dynamically adjust decision boundaries.
- Applied statistical modeling (log-normal RMSE, z-score thresholding) to dynamically detect anomalies in streaming data, improving detection reliability under changing network conditions.
- Tuned model parameters (z, window size) and evaluated performance on 700K+ packets (Mirai dataset) using latency, alert rate, and throughput metrics.
- Developed visualizations to analyze threshold evolution, anomaly scores, and system performance in streaming environments.

Adventure Works End-to-End Data Engineering Pipeline | Azure, ADF, Databricks, Synapse **Nov 2025 - Jan 2026**

Adventure Works End-to-End Data Engineering (Azure)

- Built a secure, end-to-end Azure Medallion pipeline on Linux VMs: ingested multi-source data into ADLS with encryption at rest, transformed via Databricks PySpark, and served curated datasets in Synapse for Power BI business dashboards
- Enhanced metadata-driven ingestion in ADF with robust error handling and logging: Lookup reads JSON configs, ForEach executes parameterized Copy activities into ADLS Bronze with retry logic and detailed activity logs
- Developed Databricks PySpark transformations and wrote optimized Parquet (Snappy) outputs to ADLS Silver (Calendar Month/Year, cleaned ProductSKU/ProductName, enhanced Sales fields).

Education

Texas State University **Jan 2025 - Dec 2026**
Master of Science, Computer Science (GPA: 4.0) *San Marcos, TX*

- **Coursework:** Advanced Data Mining, Database Theory and Design, Machine Learning & Essentials

GITAM Deemed University **Jun 2018 - Jul 2022**
Bachelor of Technology, Computer Science (GPA: 3.5) *India*

Certifications

- Building AI Multimodal Systems: NVIDIA
- Python for Data Science & Development: IBM
- Databases & SQL for Data Science: IBM